GLS-enabled index calculus at $k = 4$: the GLS endomorphism on the V_4 line (E25)

`RESEARCH_GLV_INVARIANT_FACTOR_BASES.md` §9.15 (registered before code) and §9.16 · 2026-10-18

1. Result

On GLS twists over $\mathbb{F}_{p^4}$ of curves over $\mathbb{F}_{p^2}$ whose 2-torsion the twist splits, the GLS endomorphism ψ ($\psi^2 = -1$) keeps E24′s V_4 base line and acts on it as $z \mapsto \pm i/z$, so the base folds by $\langle V_4, \psi \rangle$ with 8 points a column instead of 4. Measured end to end on one curve, one base and one target stream (10 curves, $p = 101$–397), **the ψ fold cuts the $k = 4$ index calculus by 1.35–2.67 × (median 1.86 ×), and against the matched $\langle \pm\psi \rangle$ walk the ψ arm gains 1.26–1.97 × over the control at every $p \geq 197$.** The modelled handover on these curves moves from $2^{93.2}$ (control) to $2^{88.0}$. Every logarithm on both arms and every one of the 960 walks verified. **Class: engineering, a constant** — the fitted exponent of $S/\sqrt{\text{rho}}$ is -0.29 against -0.30.

Distance remaining. Every row is still $1.1 \cdot 10^4$–$2.7 \cdot 10^5$ × slower than the matched rho ($1.13 \cdot 10^4$ × at $p = 397$, the best). **What it does not do.** Every curve of this construction has cofactor $h = 16$ (E24′s had 4), which makes rho 2 × cheaper at a given p and absorbs most of the fold at matched n: arm B's own handover $2^{88.0}$ against E24′s own $2^{90.1}$, from different curve families. **Scope:** $p \equiv 1 \pmod 4$, $p \leq 397$, $n \leq 2^{30.5}$.

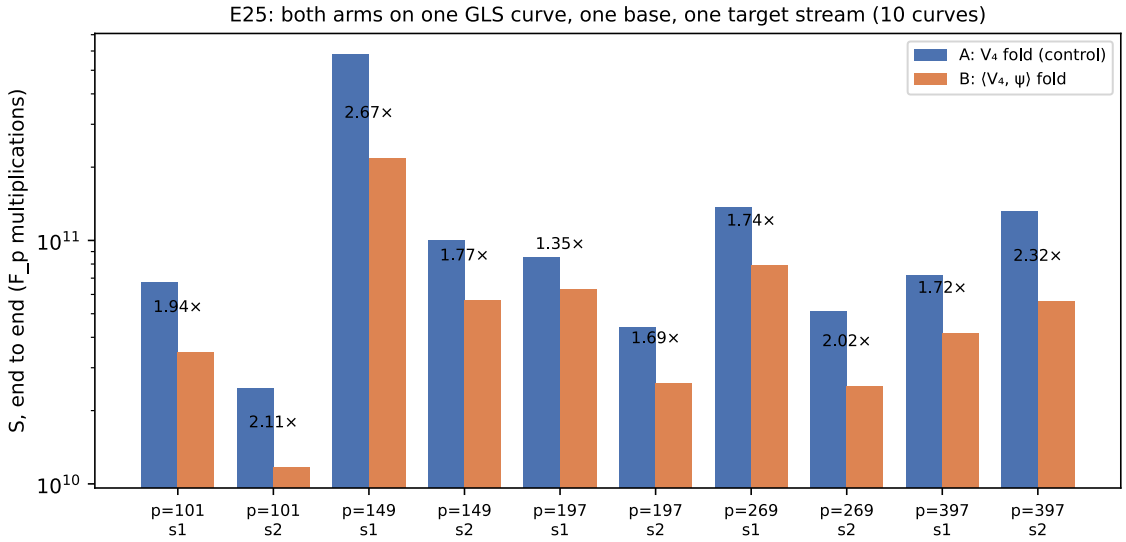


Figure 1: S end to end per curve, control (A, V_4 fold) and ψ arm (B, $\langle V_4, \psi \rangle$ fold), same curve, base and target stream; run `experiments/30_gls_v4` (binary `46202dc97`).

2. What ran

Registered in §9.15 before any code. Curves: `generate_gls_klein_instance` — $e_2 \in \mathbb{F}_{p^4} \setminus \mathbb{F}_{p^2}$, $e_3 = e_2^{p^2}$, $e_1 = -e_2 - e_3$, roots μe_i with μ a non-square; the GLS order $(p^2 - 1)^2 + t^2$, $\psi^2 = -1$, $\psi(G) = [\lambda]G$, ψ fixing T_1 and swapping T_2, T_3 all checked on points. Arms: E24′s pipeline unchanged (the V_4-invariant S_5 solve at weighted degree 11, one verified decomposition per target, Wiedemann until $[d]G = Q$ verifies, every phase charged in $\mathbb{F}_p$ multiplications), with the base folded by V_4 (A) or $\langle V_4, \psi \rangle$ (B, coefficients $\pm\lambda^e$ each checked on points). Walks: 16 each of $\langle \pm\psi \rangle$, negation and unfolded per curve. Host: one shared 4-vCPU container; operation counts only.

Stage A (gate), 56 curves, $p = 29$–397: every curve exists; 8.00 points a column on every non-empty base; $\nu^2 \notin \mathbb{F}_p$ on all; distinct column logarithms equal the column count wherever computed ($p \leq 149$): not confined. Density 0.49 pooled, 0–0.75 per curve (orbits of 8 lift together). $h = 16$ on every curve.

p	seed	$\log_2 n$	cols A/B	S_A/S_B	neg/$\pm\psi$ steps	$S_A/\sqrt{\text{rho neg}}$	$S_B/\sqrt{\text{rho}\ \pm\psi}$	impr.
101	1	22.6	10/5	1.94	1.03	$1.53 \cdot 10^5$	$6.77 \cdot 10^4$	2.26
101	2	22.6	14/7	2.11	1.49	$6.02 \cdot 10^4$	$2.93 \cdot 10^4$	2.05
149	1	24.9	10/5	2.67	1.09	$8.31 \cdot 10^5$	$2.65 \cdot 10^5$	3.13

149	2	24.9	14/7	1.77	0.96	$1.58 \cdot 10^5$	$7.01 \cdot 10^4$	2.25
197	1	26.5	22/11	1.35	1.52	$7.71 \cdot 10^4$	$6.11 \cdot 10^4$	1.26
197	2	26.5	30/15	1.69	1.62	$3.97 \cdot 10^4$	$2.62 \cdot 10^4$	1.51
269	1	28.3	30/15	1.74	1.61	$7.45 \cdot 10^4$	$4.84 \cdot 10^4$	1.54
269	2	28.3	40/20	2.02	2.08	$2.24 \cdot 10^4$	$1.56 \cdot 10^4$	1.44
397	1	30.5	60/30	1.72	1.36	$1.99 \cdot 10^4$	$1.13 \cdot 10^4$	1.76
397	2	30.5	48/24	2.32	1.67	$3.38 \cdot 10^4$	$1.71 \cdot 10^4$	1.97

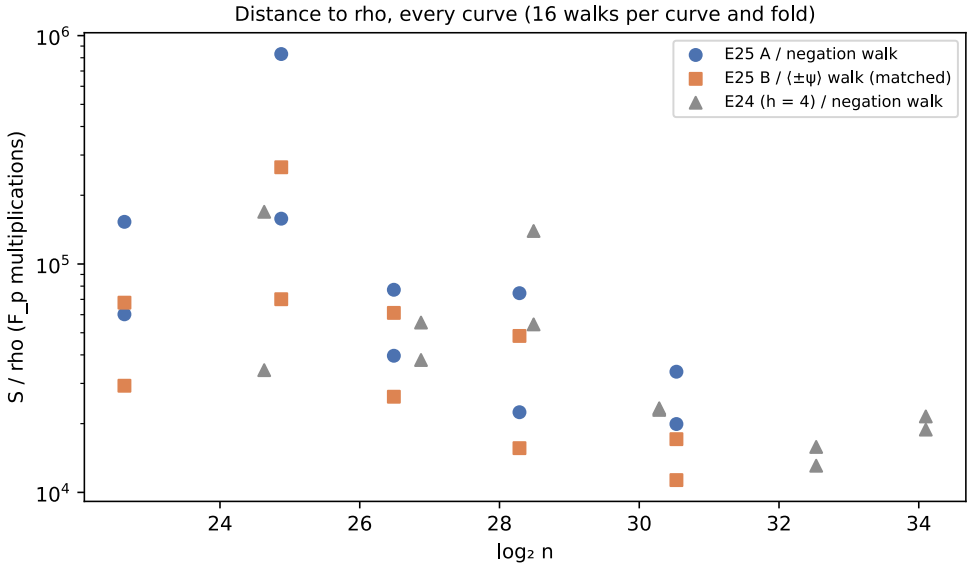


Figure 2: $S/$ rho against $\log_2 n$: E25 arm A against the negation walk, arm B against its matched $\langle \pm\psi \rangle$ walk, and E24's 12 curves ($h = 4$) against the negation walk; 16 walks per curve and fold.

3. Predictions and verdicts

id	registered	measured
G2.32	curves exist, ψ verified, base ψ-stable, ≥ 7.8 pts/col, density $[0.3, 0.7]$, true coefficients, log recovered	holds; density 0.49 pooled, 14 of 56 curves outside per curve (the band did not say which)
G2.33	cols A/B in $[1.9, 2.0]$	2.00 on every curve: holds
G2.34	y_B/y_A in $[0.8, 1.25]$	0.69–1.42, pooled 0.91: falsified per curve (5–60 relations a curve)
G2.35	S_A/S_B in $[1.6, 2.4]$	1.35–2.67, median 1.86: falsified per curve (8 of 10 in band)
G2.36	steps mean $[1.2, 1.6]$; improvement $[1.1, 2.0]$	steps 1.44 holds; improvement in band at $p \geq 197$, above it at $p \leq 149$, where walk set-up dominates
G2.37	$n^* \in [2^{84}, 2^{92}]$	$2^{88.0}$ ($2^{85.7}$ from $p = 397$): holds

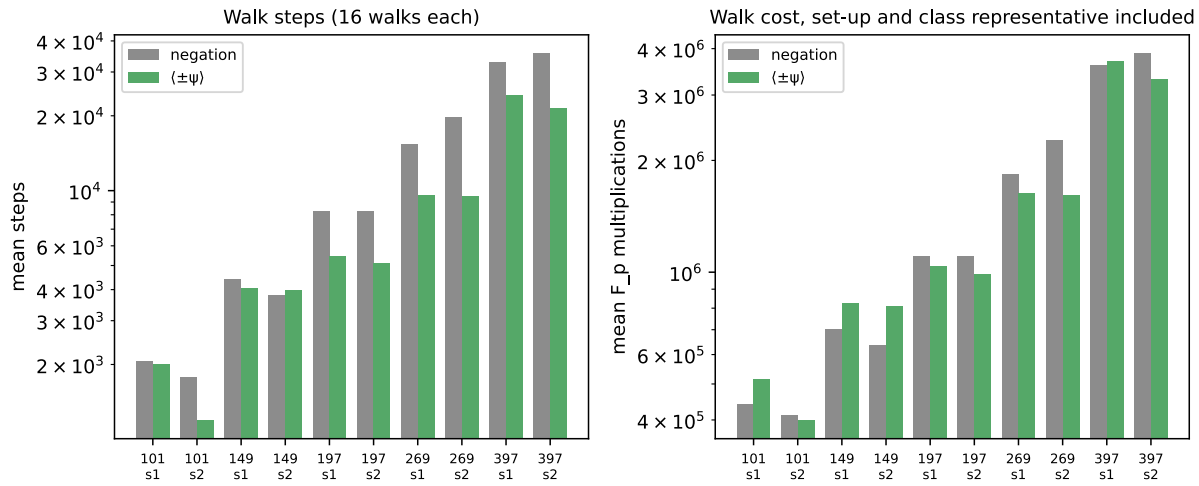


Figure 3: The matched walks: steps (left) and $\mathbb{F}_p$ multiplications with set-up and class representatives (right), negation against $\langle \pm\psi \rangle$, 16 walks each per curve. Below $n \approx 2^{25}$ set-up dominates and the ψ walk costs more.

4. Addendum: a GLS construction with $h = 4$ (E26)

Registered in §9.17 before code; results §9.18. On the GLS twist, $E(\mathbb{F}_{q^2})$ is a $\mathbb{Z}[i]$-module killed by $(q-1) - t\psi$, so its 2-part is $(\mathbb{Z}/2)^2$ exactly when $t \equiv 2 \pmod 4$. The V_4 line needs the halves of T_1 and T_2 to have rational abscissae; when $h = 4$ that forces $E_0[4] \subset E_0(\mathbb{F}_{q^2})$, i.e. $\pi^2 \equiv 1 \pmod 4$ for the q-Frobenius of E_0, which is impossible when π swaps T_2, T_3 and $\det \pi = q \equiv 1 \pmod 4$. **Census of 560 twists, $p = 29$–397:** the 310 curves with $t \equiv 2 \pmod 4$ all have $v_2(\#E) = 2$ and all fail the halves condition; the 250 with $4 \mid t$ all satisfy it and all have $v_2(\#E) \geq 4$. **No GLS twist carries the V_4 line with $h = 4$; E25′s $h = 16$ is the minimum** (accounting; records `experiments/31_gls_cofactor/`).

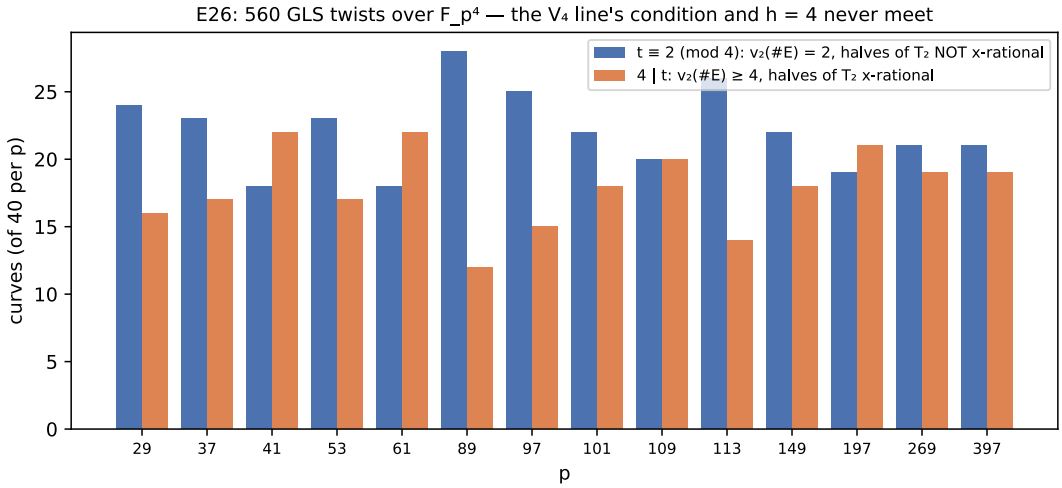


Figure 4: E26: per p, the 40 GLS twists split into the two classes; the V_4 line's condition holds exactly on the $4 \mid t$ class.

5. Records

Raw (E26): `experiments/31_gls_cofactor/census.json`. Raw (E25): `experiments/30_gls_v4/p{101,149,197,269,397}s{1,2}.json` (both arms in each), `stage_a.json`, `run.log`, `tables.md`. Code: `src/cryptanalysis/quartic_folds.rs` (`generate_gls_klein_instance`, `klein_base_with`, `RhoFold::Gls`, `run_k4_gls_pipeline`, `run_gls_stage_a`), `examples/quartic_pipeline_gls.rs`, `examples/gls_stage_a.rs`. Figures: `figures/plot.py` (recorded values only).